



HALO:

High-efficiency Autonomous Low-SWaP Operations

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MILESTONE 4 PROGRESS MATRIX

Task	Completion %	Sloan	To Do
Implement FP16 engine from FP32 ONNX model	100%	Compile the FP32 ONNX model into a TensorRT engine using FP16	None
Implement INT8 engine	100%	Build an INT8 engine using the TensorRT framework from the Jetson SDK	None
Write an Inference Script for the INT8 Model	100%	Write a Python inference script for the INT8 engine	None
Record Metrics for 8-bit Model	100%	Get inference metrics from the INT8	None

TASKS COMPLETED

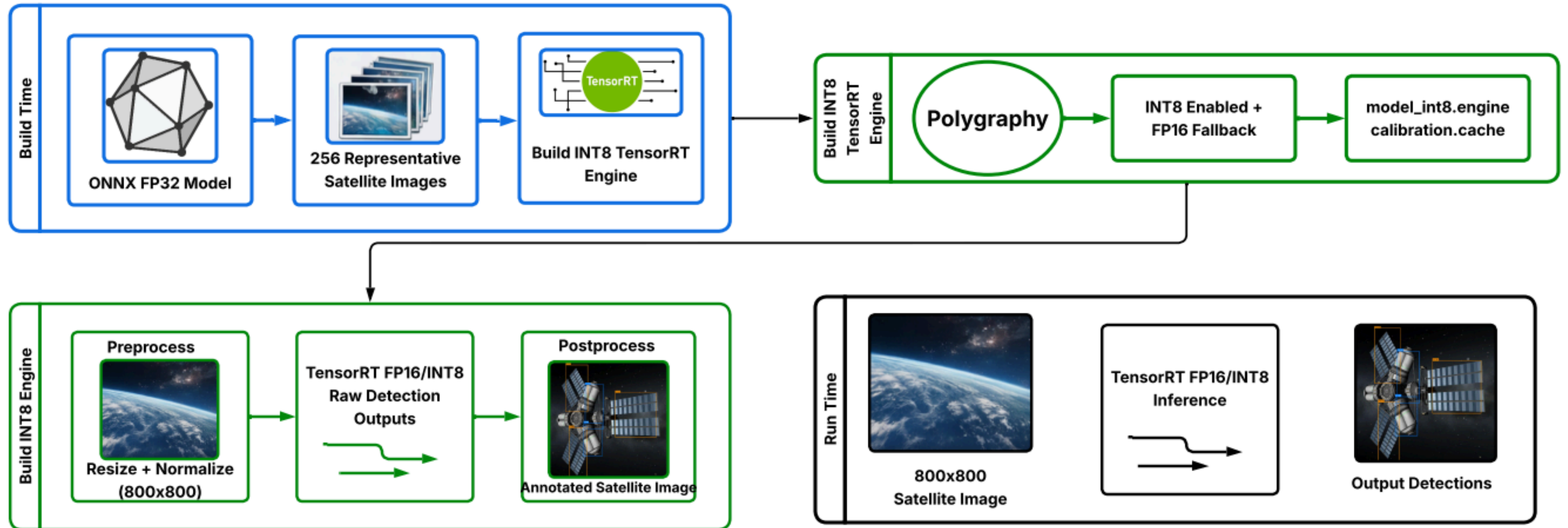
- Implement FP16 engine from FP32 ONNX model
 - Used TensorRT to compile FP32 ONNX into a device-specific engine for FP16 engine using trtexec
- Implement an INT8 Pipeline on Jetson Orin
 - Utilized TensorRT's Polygraphy tools to run calibration on 256 representative images
- Write an Inference Script for the INT8 Model
 - Produces detections and draws bounding boxes
- Compute/Record Metrics for 8-bit Model
 - Get mAP (mean Average Precision) scores for each model

MODEL COMPARISON

mAP@[0.50:0.95]: FP32 0.0958 → FP16 0.0724 → INT8 0.0735

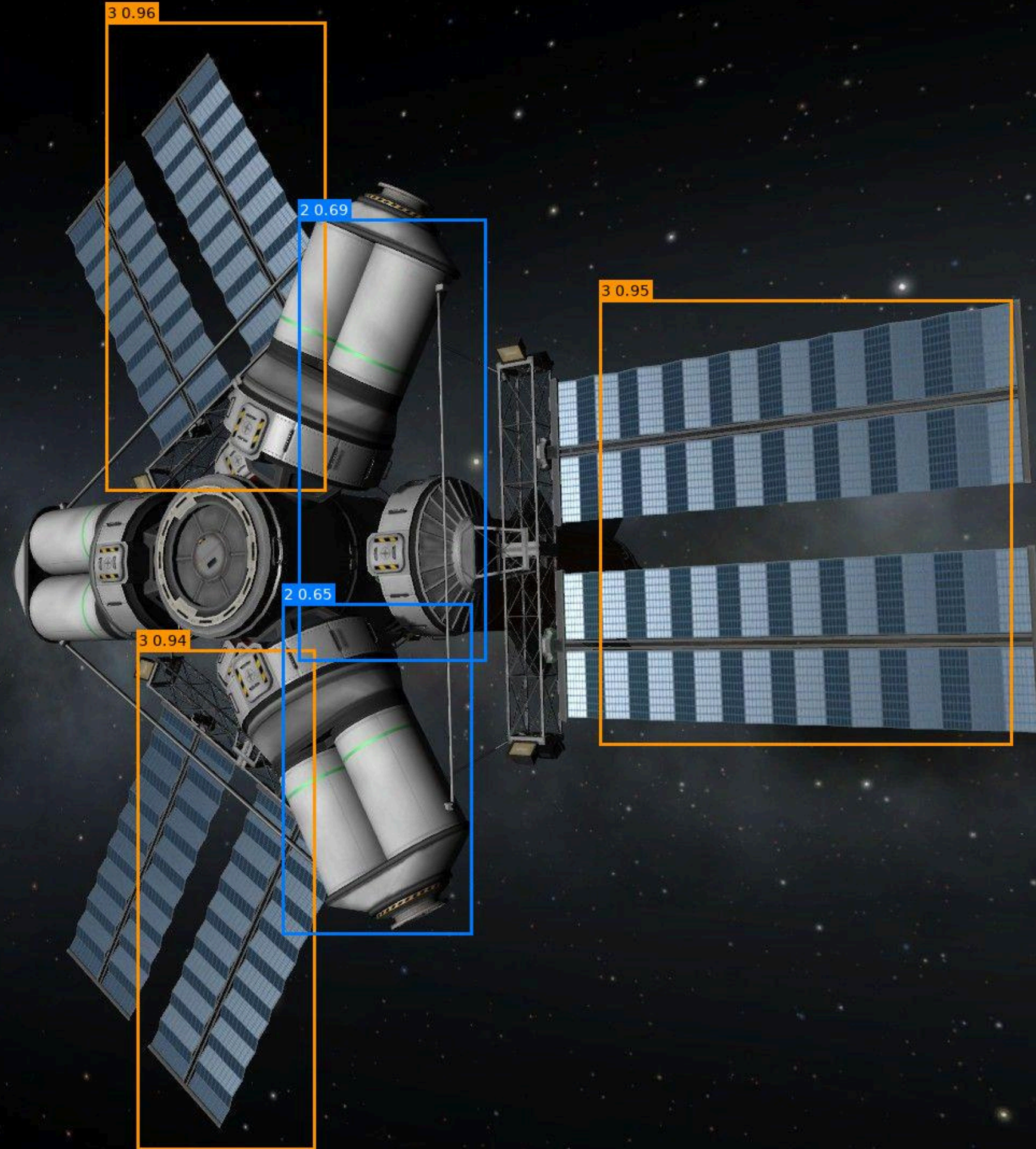
- FP32 → FP16 $\approx$ -24% drop
 - due to precision/runtime differences introduced by TensorRT
 - Solutions:
 - Ensure FP16 TensorRT uses identical preprocessing and postprocessing as FP32
 - Quantized Aware Training/Retraining
- FP16 → INT8 $\approx$ 0% drop
 - INT8 used calibration to estimate the typical activation ranges and set quantization scales (how real valued activations are mapped to INT8) → Minimizes clipping → helps keep INT8 metrics close to FP16

INT8 PIPELINE



DEMO

■ 2: 2
■ 3: 3



MILESTONE 5 TASK MATRIX

Task	Sloan
Implement 4-bit Model	100%
Begin progress on 2-bit Model	100%
Formal Compilation of Results thus far for Publication	100%

MILESTONE 4 TASKS

- Implement 4-bit Model
 - Run INT4 inference engine using TensorRT
- Begin Progress on 2-bit Model
 - No specific support for INT2 inference engine, implement manually
- Formal Compilation of Results
 - Faculty advisor's request, compile results thus far to begin publication



THANK YOU!
QUESTIONS?